

Actify DevOps Engineer Assessment

Candidate Details

Name: Rupesh Sokande

Date: 02 June 2026

GitHub Repository:

<https://github.com/rupeshsokande10-eng/actify-devops-assessment>

1. Objective

The objective of this assessment was to deploy a static website on an Azure Virtual Machine, configure NGINX as a web server and reverse proxy, deploy a backend application using Docker, and implement a CI/CD pipeline using Jenkins integrated with GitHub.

2. Azure Infrastructure Setup

Cloud Platform

Microsoft Azure

Virtual Machine Details

Parameter	Value
VM Name	actify-vm
Operating System	Ubuntu Server 24.04 LTS
Resource Group	actify-rg
Public IP	4.240.95.205

Screenshot

Dashboard > Compute infrastructure | Virtual machines

actify-vm
Virtual machine

Help me copy this VM in any region | Manage this VM with Azure CLI

Search

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Monitor
- Resource visualizer
- Connect
- Networking
 - Network settings
 - Load balancing
 - Application security groups
 - Network manager
- Settings
- Availability + scale
- Security

Connect | Start | Restart | Stop | Hibernate | Capture | Delete | Refresh | Scale | Open in mobile | Feedback

Essentials

JSON View

Resource group (move)	actify-rg	Operating system	Linux (ubuntu 24.04)
Status	Running	Size	Standard B2s v2 (2 vcpus, 8 GiB memory)
Location	Central India (Zone 1)	Primary NIC public IP	4.240.95.205
Subscription (move)	Azure subscription 1	1 associated public IPs	
Subscription ID	fae5d6ce-e57f-4a31-86b4-85a61642cc4a	Virtual network/subnet	actify-vm-vnet/default
Availability zone	1	DNS name	Not configured
		Health state	-
		Time created	2/6/2026, 9:11 am UTC

Tags ([edit](#))
[Add tags](#)

Properties | Monitoring | Capabilities (7) | Recommendations | Tutorials

Screenshot

Dashboard > Compute infrastructure | Virtual machines > actify-vm

actify-vm | Network settings

Virtual machine

List all my network interfaces for this VM | How can I make this VM secure? +1

Search

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Network security group **actify-vm-nsg** (attached to networkInterface: **actify-vm36_z1**)
Impacts 0 subnets, 1 network interfaces

+ Create port rule

Search rules

Source == all | Destination == all | Protocol == all | Action == all | Port == all

Prio...	Name	Port	Protocol	Source	Destination	Action
Inbound port rules (7)						
300	SSH	22	TCP	Any	Any	Allow
320	HTTP	80	TCP	Any	Any	Allow
340	HTTPS	443	TCP	Any	Any	Allow
350	jenkins	8080	TCP	Any	Any	Allow
65000	AllowVnetInBound	Any	Any	VirtualNetwork	VirtualNetwork	Allow
65001	AllowAzureLoadBalancerInB...	Any	Any	AzureLoadBalancer	Any	Allow
65500	DenyAllInBound	Any	Any	Any	Any	Deny
Outbound port rules (3)						

3. SSH Access

Successfully connected to the Azure Virtual Machine using SSH from Windows PowerShell.

Command Used:

```
ssh azureuser@4.240.95.205
```

Screenshot

```
PS D:\actify-devops-assessment> ssh azureuser@4.240.95.205
azureuser@4.240.95.205's password:
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1017-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jun  2 11:33:12 UTC 2026

System load:  0.04          Processes:           156
Usage of /:   11.8% of 28.02GB Users logged in:        0
Memory usage: 16%          IPv4 address for eth0: 10.0.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

1 additional security update can be applied with ESM Apps.
Learn more about enabling ESM Apps service at https://ubuntu.com/esm

Last login: Tue Jun  2 10:59:03 2026 from 103.158.146.122
azureuser@actify-vm:~$
```

4. Source Code Management

The application source code was managed using Git and GitHub.

Repository Structure:

actify-devops-assessment

├─ Jenkinsfile

├─ README.md

├─ nginx

| └─ actify.conf

└─ website

└─ index.html

Screenshot

The screenshot shows the GitHub interface for the repository 'actify-devops-assessment'. At the top, the user 'rupeshsokande10-eng' is logged in. The repository is public and has 1 branch (main) and 0 tags. The commit history shows 8 commits. The file list includes 'nginx' (Add NGINX configuration, yesterday), 'website' (Initial commit, yesterday), 'Jenkinsfile' (Fix Jenkins pipeline, 22 minutes ago), and 'README.md' (Initial commit, yesterday). The README section is visible, showing the repository name 'actify-devops-assessment'.

5. NGINX Configuration

NGINX was installed and configured to serve the static website and reverse proxy requests to the backend Docker application.

NGINX Configuration:

```
server {  
    listen 80;  
  
    location / {  
        root /var/www/html/site;  
  
        index index.html;  
    }  
  
    location /app/ {  
        proxy_pass http://127.0.0.1:3000/;  
    }  
}
```

Screenshot

```
azureuser@actify-vm:~$ cat /etc/nginx/sites-available/actify
server {
    listen 80;

    location / {
        root /var/www/html/site;
        index index.html;
    }

    location /app/ {
        proxy_pass http://127.0.0.1:3000/;
    }
}
```

Screenshot

```
azureuser@actify-vm:~$ sudo nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
```

6. Website Deployment

The static website was deployed to:

<http://4.240.95.205>

Result:

Website successfully accessible through NGINX.

Screenshot

Welcome to Actify DevOps Assessment

Deployed using Jenkins CI/CD Pipeline

7. Docker Deployment

A backend application was deployed using Docker.

Docker Image Used:

nginxdemos/hello

Container Run Command:

```
sudo docker run -d --name backend -p 3000:80 nginxdemos/hello
```

Verification Command:

```
sudo docker ps
```

Result:

Docker container running successfully.

Screenshot

```
azureuser@actify-vm:~/actify-devops-assessment$ sudo docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
2fcd3de1c295   nginxdemos/hello  "/docker-entrypoint...."  43 seconds ago  Up 42 seconds  0.0.0.0:3000->80/tcp, [::]
:3000->80/tcp   backend
azureuser@actify-vm:~/actify-devops-assessment$ curl http://localhost:3000
<!DOCTYPE html>
<html>
<head>
<title>Hello World</title>
<link href="data:image/png;base64,iVBORw0KGgoAAAANSUhEUgAAAAEAAAABAAQCAQAACgAAAAEgAAQAAAQAAAABAAAAAABAAQAAAEAAABACAYAAACQaXHeAAAGPElEQVR42u1bDUyUdRj/iwpoIMcbZqtXfNn
suScez/OIMg1V7SFONuaU8P1Mwy1lcPUyHk1uVbKcXfvy6GikTGKmpEyoiejJipouUBcgsinhwUKKKJ8PD3vnzsxuLv35Q644+Ue9mmwH3P3f5/d7n6/3/3
+OEJ/4xc8+BYQtQuJwB0kIp+JrzUTB7iJuweBf4baTlJ5oCqw11C/JHp+tnqBb1ngT4z8WgReTUGbWCBGq0qvKRFcHF4eT/ZFBKoLVMBGibhiYkaQIjca
fLAK+D8z9YhjXmgsVUGc84+gyx9AYD0khXcmFLCmUBL68HMZ+PnHxyFw3Uwi8B8hgJYh7j4c7c8PV5CEbUTUzBoHcU78i1l/FYFXWmPaNec3q4mz5YcqJP
I1JGKqL2Z3hkcjD5EUznmcu6qiNT+Y2CPEoH3Wm4A/QERWQFe9QQ0caeCD1SZJrht1HxG0D3sOuCEiCA1aj4ZY3Ipz18LiVtn8hxi5ZRgWM8YYPBODF/9z
xOLcVRRVs+YgtwFzxcS1Bo9y+avBiOTQeUzWi3F5+kOwxSxKkmWNNHrjUokqtqtSyysw5gUHV4mtmZEHSdRkl+aELvcFIRN397gPPXD4ZgbxJW1SSOJD660
MgUayHu1KfAz+pfCUTwr+HUQc80RQ1jK4ZgGsTvcY5uQP5oYkY2Hfck5sGLpS611xZQmNn7Xkedp3QgMrWC1DX0Qwnms/A1rK9cF9atNVo18DP/3o5fF99
BGo7LFDRIWgMJQaYQv/PyOchySP0TITrBIHyb+WShLrLNGEx5NeXgj2paW8C5rs46h3Dc3kt3G20gr9aqoes+f5Rvbl1aJ5iXnKnxkfIEoB3N/zHeHAmF9
ovwryvYvC9TysnICkEonPX212vvOU8+As6eS+QCDAw0aNLABq6L08DkJMSSznMMEfScFFGwCJYXbDV7lq17RYIQu+QTYpjRUBM3gZQIT+cOwyTpwRpyBQR
sKrgU4ceIS4JkCsXlI1+ZsIS0NVXB6sLE/tL5EQkQJkOm52YON9y7glqJkCS0qzrD6Uvc1wZ1EBA07V/IafmN4ckHG+ugJkSEHUvQQ0ENfy9BLP3R0NR4y
mHJGRwFWBnZ6fPVwMBF9EDgrD2z0USqtoaHJKw49SBoZ2dwggTxmCEsvspYLLi4PKNDrvv68OfuKlt/68MqiJAan4Q0IpDm6G7r8fue692X4fI7P1ByqA6
AqygNh0XHiACld0kpz9aGVRJAB08CTP+3sqfHZJQeqkSgvHZn+xaqEICKAlhECsG060MwdVF4IcesDL/ExUSYN3okCrD31fqHZLwclqk5owPVUoA3UcIgd
Bv10BrV7vdz3b39kBhw0kVE2BNirG/bqRghyPqIcBKQkKJcVgE1LQ1wR3S50oqCDBK1SEUzGdyFBNwvq1RTQT0b4BOF5+BgoayCUqATLMSXsRz16uHX8E
ON0UtXS2KcFaus0syvWFLV1tznNAuzfLAGxb+R/esGuodDcd0bUvBYLe1hrf/mWd08ogdyTtjNwYbIsrORhBIwJMPOTWHL1i6Lriz107FUKviivcZvfp8W
ZvN8TmbVS2rtSH8mMtn9gSe50KAz79yLw8490OGYpp81sTUGictd3EA6PHWwB20+mYUNUro/aMs4dhqjsdco0WgXh5yYU0g0P0EzFBd7DxZoVHY7aHmWt
B6VunwhLB6P0gFULk6zhJnnBw5HW9D9N5GkpQEjMBcQ0g+JMBNxcjMZgHISawvGZHiKw+0mybv5ozP0txgvk07AQvWxAoh98sXsur3RmwMstxIud9fiIzM
AIXTV6yNqxHaH7gg1GA7bgxVvHfEjQ1hA110ZM/A46g00x0b0PoiHpSEdvsMZhXVbvVRL4TLz2E140EK1dgsnnd9mBaHcmwuigJHeCGLkXvHNANHOBP4J/
HYmoGBGwsJU1ka0nAvM2ht40758ZnmvVRRJ2413roMa7MxVq4jPrdyMRc8bh9wR0TyIRWR9hzNXa3s3Ftif6KDWuBcBH0hErky2bnrAv5E9jcbjiapE1E
xHk08iEY10vjLTjAkugezh7ySqFUP0XHTtZAR7ncY4rRrYYgtcCtGHPUGmjehPmiKXjXc/14g6HfGJT3ziEw/If86JzB/YMku9AAAAAE1FTkSuQmCC" re
```

Ln 10, Col 8 - Sp

8. NGINX Reverse Proxy

The backend application was exposed through NGINX reverse proxy.

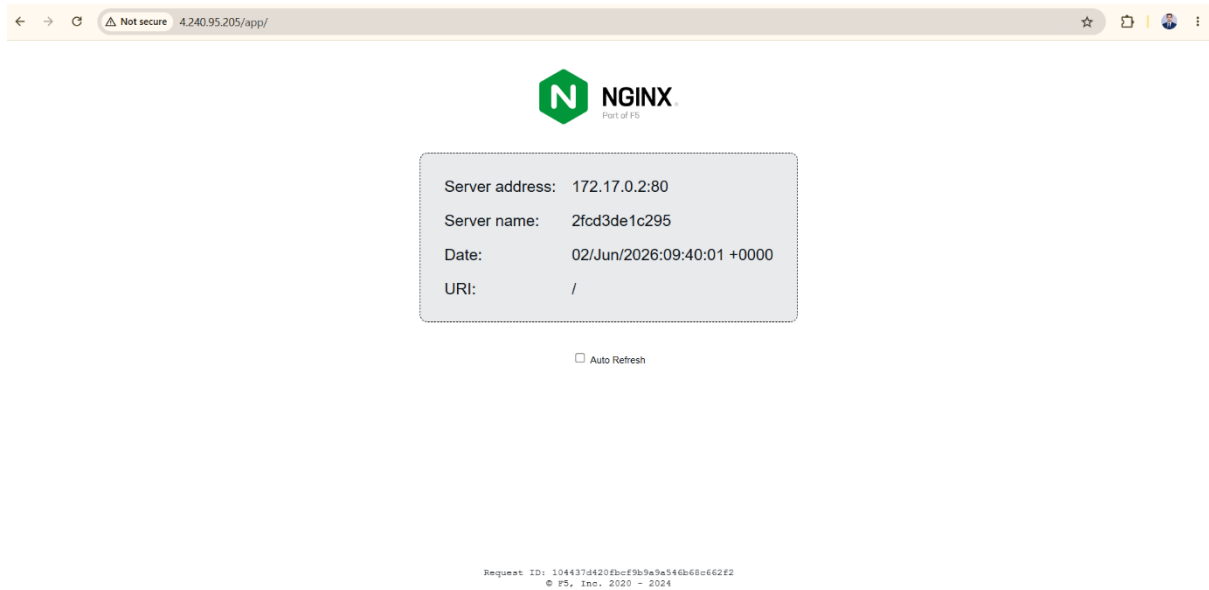
URL:

<http://4.240.95.205/app/>

Result:

Backend Docker application accessible through browser.

Screenshot



9. Jenkins CI/CD Pipeline

Jenkins was installed on the Azure VM and integrated with GitHub.

Jenkins URL:

<http://4.240.95.205:8080>

Pipeline Stages:

1. Checkout
2. Deploy
3. Verify

Jenkinsfile:

```
pipeline {
  agent any

  stages {

    stage('Checkout') {
      steps {
        git branch: 'main',
          url: 'https://github.com/rupeshsokande10-eng/actify-devops-assessment.git'
```



```

    }
}

stage('Deploy') {
    steps {
        echo 'Deploying website'
    }
}

stage('Verify') {
    steps {
        echo 'Deployment successful'
    }
}
}
}

```

Screenshot

The screenshot shows the Jenkins Dashboard interface. The top navigation bar includes the Jenkins logo, a search icon, and user information for 'Rupesh Sokande'. The main content area displays a table of build history for the 'Actify-Pipeline'.

S	W	Name	Last Success	Last Failure	Last Duration
✓	☁	Actify-Pipeline	38 min #2	43 min #1	3.3 sec

Below the table, there is a 'Build Queue' section showing 'No builds in the queue.' and a 'Build Executor Status' section showing '0/2'.

Screenshot

Dashboard > Actify-Pipeline > Configuration

Configure

Pipeline script from SCM

SCM ?

Git ?

Repositories ?

Repository URL ?

https://github.com/rupesthokande10-eng/actify-devops-assessment.git

Credentials ?

- none -

+ Add

Advanced

Add Repository

Branches to build ?

Branch Specifier (blank for 'any') ?

*/main

Save Apply

Screenshot

Commit message: "Fix Jenkins pipeline"

```
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Deploy)
[Pipeline] echo
Deploying website
[Pipeline] }
[Pipeline] // stage
[Pipeline] stage
[Pipeline] { (Verify)
[Pipeline] echo
Deployment successful
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // withEnv
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS
```

10. Troubleshooting Findings

Scenario 1: NGINX Service Stopped

Investigation:

```
sudo systemctl status nginx
```

Finding:

NGINX service was stopped.

Resolution:

```
sudo systemctl start nginx
```

Scenario 2: Azure NSG Configuration Verification

Investigation:

Website accessibility was verified from the browser.

Finding:

Reviewed Azure Network Security Group (NSG) inbound rules to ensure required ports were open.

Resolution:

Verified and confirmed that inbound TCP ports 22 (SSH), 80 (HTTP), 443 (HTTPS), and 8080 (Jenkins) were allowed in the NSG configuration.

Scenario 3: Incorrect NGINX Configuration

Investigation:

```
sudo nginx -t
```

Finding:

Syntax error in NGINX configuration.

Resolution:

Corrected configuration and reloaded NGINX.

Scenario 4: Backend Application Not Running

Investigation:

```
sudo docker ps
```

Finding:

Backend container was not running.

Resolution:

`sudo docker start backend`

11. Conclusion

Successfully completed the following tasks:

- Azure VM Provisioning
- GitHub Repository Management
- NGINX Installation and Configuration
- Static Website Deployment
- Docker Backend Deployment
- NGINX Reverse Proxy Configuration
- Jenkins CI/CD Pipeline Setup
- Successful Automated Build Execution

All mandatory requirements of the Actify DevOps Engineer Assessment were completed successfully.